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Database Systems (CSF212)

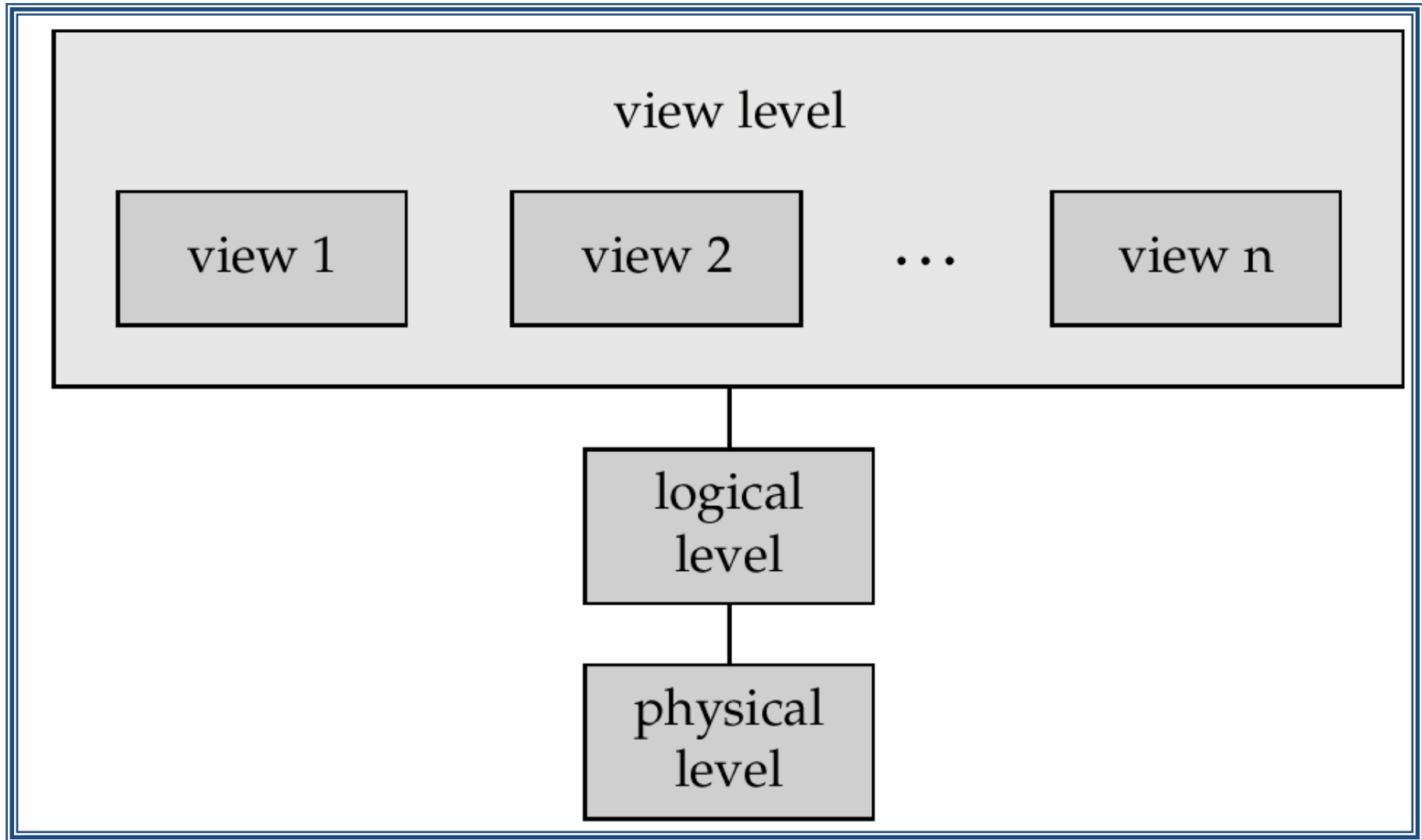
by

Dr. Shubhangi

Dept. of CS and IS



View of Data



Views

- In some cases, it is not desirable for all users to see the entire logical model (i.e., all the actual relations stored in the database.)
- Consider a person who needs to know a customer's loan number but has no need to see the loan amount. This person should see a relation described, in the relational algebra, by $\Pi_{customer-name, loan-number} (borrower \bowtie loan)$
- Any relation that is not of the conceptual model but is made visible to a user as a “virtual relation” is called a **view**.

View Definition

- A view is defined using the **create view** statement which has the form

create view v as <query expression>

where <query expression> is any legal relational algebra query expression. The view name is represented by v .

- Once a view is defined, the view name can be used to refer to the virtual relation that the view generates.
- View definition is not the same as creating a new relation by evaluating the query expression
 - Rather, a view definition causes the saving of an expression; the expression is substituted into queries using the view.

Views

To provide a mechanism to hide certain data from the view of certain users.

Eg. Create view all-cust as

(select branchname, custname

From depositor, account

Where depositor.account no = account.account no)

union

(select branchname, custname

From borrower, loan

Where borrower.loan no = loan.loan no)

Find all customers of Zuarinagar branch

Select custname

From all-cust

Where branch-name = 'Zuarinagar';

View Examples

- Consider the view (named *all-customer*) consisting of branches and their customers.

create view *all-customer* **as**

$$\begin{aligned} &\Pi_{branch-name, customer-name} (depositor \bowtie account) \\ &\cup \Pi_{branch-name, customer-name} (borrower \bowtie loan) \end{aligned}$$

We can find all customers of the Zuarinagar branch by writing:

$$\begin{aligned} &\Pi_{customer-name} \\ &(\sigma_{branch-name = \text{"Zuarinagar"}} (all-customer)) \end{aligned}$$

Updates Through View

- Database modifications expressed as views must be translated to modifications of the actual relations in the database.
- Any updates made in views are reflected in its base table also.

This is the diff betn rename and view.

- Consider the person who needs to see all loan data in the *loan* relation except *amount*. The view given to the person, *branch-loan*, is defined as:

create view *branch-loan* as

$\Pi_{branch-name, loan-number}(loan)$

- Since we allow a view name to appear wherever a relation name is allowed, the person may write:

$branch-loan \leftarrow branch-loan \cup \{("Zuarinagar", L-37)\}$

Updates Through Views (Cont.)

- The previous insertion must be represented by an insertion into the actual relation *loan* from which the view *branch-loan* is constructed.
- An insertion into *loan* requires a value for *amount*. The insertion can be dealt with by either.
 - rejecting the insertion and returning an error message to the user.
 - inserting a tuple (“L-37”, “Sancoale”, *null*) into the *loan* relation
- Some updates through views are impossible to translate into database relation updates
 - create view *v* as $\sigma_{branch-name = \text{“Sancoale”}}(account)$
 $v \leftarrow v \cup (L-99, Madgaon, 23)$
- Others cannot be translated uniquely
 - $all-customer \leftarrow all-customer \cup \{(\text{“Zuarinagar”, “John”})\}$
 - Have to choose loan or account, and create a new loan/account number!